

→ GIT is widely used due to:

- (1) Collaboration → multiple team members can work on the same scripts (eg python, SQL or notebooks)
- (2) Safety → It acts like a "time machine"
- (3) Experimentation → you can try out a new analysis or feature in isolation without messing up the stable version of the project.
- (4) Auditability → Every change is recorded with a reason (the commit message), showing who changed what and why

→ A commit - is a "snapshot" of your work at a specific moment in time, when you commit, you are essentially saving a permanent milestone in the project history with a descriptive message explaining what was done.

→ Branch → A separate line of development that "branches off" from the main project. It allows you to work on new features or bug fixes in an isolated environment.